

AGILE SPRINT PLANNING DOCUMENT

Group No. - 3

Project Title

LabStock AI – Intelligent Laboratory Inventory Management System

Team Members

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1. Introduction

Agile Sprint Planning is the process of dividing a software project into smaller, manageable development cycles called sprints. Each sprint has a specific goal, a set of prioritized tasks, and a defined deliverable.

For LabStock AI, Agile Sprint Planning converts the requirements identified during Requirement Analysis and the entities defined during Database Design into manageable development tasks.

The system is designed for a clinical diagnostic laboratory and focuses on inventory and procurement management. It manages laboratory supplies, stock levels, suppliers, purchase orders, inventory transactions, demand forecasting, stockout risk, and intelligent reorder recommendations.

2. Purpose of Agile Sprint Planning

- Divide the project into manageable development increments.
- Prioritize the most important features.
- Assign requirements to specific sprints.
- Define clear sprint goals and deliverables.
- Track development progress.
- Allow testing and validation after each increment.
- Reduce development risk by building the system incrementally.
- Ensure that database design and functional requirements are implemented systematically.

3. Agile Development Approach

The project follows an iterative and incremental Agile approach.

Development Flow:

Requirement Analysis → Product Backlog → Sprint Planning → Sprint Development → Testing → Sprint Review → Sprint Retrospective → Next Sprint

4. Product Vision

The vision of LabStock AI is to provide a centralized and intelligent platform for managing laboratory inventory and procurement.

- Manage laboratory supplies.
- Monitor stock levels.
- Identify low-stock supplies.
- Manage suppliers.
- Create and track purchase orders.
- Track inventory transactions.
- Maintain historical consumption data.
- Forecast future demand.
- Identify stockout risk.
- Generate intelligent reorder recommendations.

5. Product Backlog

ID	User Story / Requirement	Priority	Story Points
US01	User registration and role-based login	High	5
US02	Manage laboratory supplies	High	8
US03	Manage suppliers	High	5
US04	Monitor current stock	High	5
US05	Identify low-stock supplies	High	5
US06	Record inventory transactions	High	8
US07	Create purchase orders	High	8
US08	Manage purchase-order items	High	5
US09	Track partial order fulfillment	High	5
US10	Maintain inventory history	Medium	5
US11	Generate demand forecasts	Medium	8
US12	Identify stockout risk	Medium	5
US13	Generate reorder recommendations	Medium	8
US14	Manage recommendation status	Medium	5
US15	Dashboard and reports	Medium	8
US16	Security, validation and backup	High	5
US17	System integration and testing	High	8

6. User Stories

US01 – User Authentication

As a system user, I want to log in securely so that I can access the functions permitted for my role.

Acceptance Criteria:

- User must provide valid credentials.
- Unauthorized users must not access restricted functions.
- User role must determine available functionality.

US02 – Laboratory Supply Management

As an Inventory Manager, I want to add, update, view and manage laboratory supplies so that inventory records remain accurate.

Acceptance Criteria:

- Supply name must be provided.
- Category and unit must be recorded.
- Quantity must be valid.
- Reorder point must be defined.
- Supplier can be associated with the supply.
- Supply information should be stored successfully.

US03 – Supplier Management

As a Procurement Staff member, I want to manage supplier information so that I can efficiently handle laboratory procurement.

Acceptance Criteria:

- Supplier can be added.
- Supplier information can be viewed.
- Supplier information can be updated.
- Supplier can be marked active/inactive.

US04 – Stock Monitoring

As an Inventory Manager, I want to monitor current stock levels so that I can identify supplies that may require replenishment.

Acceptance Criteria:

- Current quantity must be available.
- Reorder point must be available.
- System compares quantity with reorder point.
- Low-stock items are clearly identified.

US05 – Inventory Transactions

As a Laboratory Staff member, I want to record consumption, restocking and stock adjustments so that inventory history remains accurate.

Acceptance Criteria:

- Transaction type must be recorded.
- Quantity must be recorded.
- Date and time must be stored.
- User performing the transaction must be recorded.

US06 – Purchase Orders

As Procurement Staff, I want to create and manage purchase orders so that laboratory supplies can be procured efficiently.

Acceptance Criteria:

- Supplier must be associated with the purchase order.
- Order date must be stored.
- Expected delivery date can be recorded.
- Purchase order status must be maintained.
- Multiple supplies can be included in an order.

US07 – Partial Order Fulfillment

As Procurement Staff, I want to record the quantity received against the quantity ordered so that partially fulfilled orders can be tracked.

Acceptance Criteria:

- Quantity ordered and quantity received must be recorded.
- The system should support partially fulfilled status.
- Remaining quantities should be traceable.

US08 – Demand Forecasting

As a Laboratory Manager, I want future demand to be forecasted using historical inventory data so that I can plan procurement in advance.

Acceptance Criteria:

- Historical inventory transactions should be available.
- Forecast should be generated for a defined period.
- Predicted quantity should be stored.
- Forecast model and generation date should be recorded.

US09 – Intelligent Reorder Recommendation

As an Inventory Manager, I want the system to recommend reorder quantities so that I can replenish supplies before stockouts occur.

Acceptance Criteria:

- Recommendation considers current quantity.
- Predicted demand and safety stock are considered.
- Pending purchase quantity is considered.
- Risk level and recommendation reason are stored.
- Recommendation status is maintained.

7. Sprint Planning

SPRINT 1 – Project Foundation and Database Setup

Sprint Goal: Create the basic application architecture, database and secure user-management system.

Sprint Backlog:

- Set up ASP.NET Core project
- Configure SQL Server
- Configure Entity Framework Core
- Configure ASP.NET Core Identity
- Create Users entity
- Implement registration/login
- Implement user roles
- Create initial database migrations
- Configure database relationships

Database Entities / Modules: Users

Expected Output: A working application with database connectivity, authentication and role-based authorization.

SPRINT 2 – Inventory and Supplier Management

Sprint Goal: Implement the core laboratory inventory and supplier modules.

Sprint Backlog:

- Create LabSupplies model
- Create Suppliers model
- Create inventory CRUD operations
- Create supplier CRUD operations
- Add supply categories and units
- Add quantity tracking
- Add reorder point and safety stock
- Associate supplies with suppliers
- Add validation
- Test inventory module

Database Entities / Modules: Suppliers, LabSupplies

Expected Output: A functional inventory and supplier management module.

SPRINT 3 – Stock Monitoring, Transactions and Procurement

Sprint Goal: Implement stock monitoring, inventory history and purchase-order functionality.

Sprint Backlog:

- Implement low-stock detection
- Implement reorder-point comparison
- Create inventory transaction functionality
- Record consumption

- Record restocking
- Record stock adjustments
- Create purchase orders
- Add purchase-order items
- Track ordered quantity
- Track received quantity
- Implement purchase-order status
- Implement partial fulfillment

Database Entities / Modules: InventoryTransactions, PurchaseOrders, PurchaseOrderItems

Expected Output: Working stock tracking, inventory history and procurement workflows.

SPRINT 4 – AI/ML Demand Forecasting and Reorder Recommendations

Sprint Goal: Implement the intelligent components of LabStock AI using historical inventory data.

Sprint Backlog:

- Prepare historical transaction data
- Extract consumption history
- Prepare forecasting dataset
- Develop demand forecasting model
- Generate predicted demand
- Calculate stockout risk
- Store forecast results
- Develop reorder recommendation logic
- Calculate recommended quantity
- Calculate risk level
- Generate recommendation reason
- Manage recommendation status

Database Entities / Modules: DemandForecasts, ReorderRecommendations

Expected Output: Demand forecasts, stockout risk and intelligent reorder recommendations.

SPRINT 5 – Dashboard, Integration, Testing and Final Release

Sprint Goal: Integrate all modules, provide reporting functionality, test the complete system and prepare the final release.

Sprint Backlog:

- Integrate inventory and suppliers
- Integrate inventory and purchase orders
- Integrate forecasting and historical transactions
- Integrate reorder recommendations
- Create dashboard
- Add low-stock dashboard
- Add procurement information
- Add forecast and risk information
- Test role permissions

- Test database constraints
- Perform integration testing
- Fix bugs
- Verify backup process
- Prepare final documentation
- Prepare final demonstration

Database Entities / Modules: All modules

Expected Output: A complete integrated LabStock AI system ready for demonstration.

8. Sprint Summary

Sprint	Goal	Main Entities / Modules	Major Deliverable
Sprint 1	Foundation	Users	Authentication & database setup
Sprint 2	Inventory	Suppliers, LabSupplies	Inventory & supplier management
Sprint 3	Procurement	InventoryTransactions, PurchaseOrders, PurchaseOrderItems	Stock tracking & procurement
Sprint 4	Intelligence	DemandForecasts, ReorderRecommendations	Forecasting & intelligent reorder
Sprint 5	Finalization	All modules	Dashboard, integration & tested system

9. Requirement-to-Sprint Mapping

Functional Requirement	Database Component	Sprint
User Management	Users	Sprint 1
Inventory Management	LabSupplies	Sprint 2
Supplier Management	Suppliers	Sprint 2
Stock Monitoring	LabSupplies + InventoryTransactions	Sprint 3
Inventory History	InventoryTransactions	Sprint 3
Purchase Order Management	PurchaseOrders + PurchaseOrderItems	Sprint 3
Partial Order Fulfillment	PurchaseOrderItems	Sprint 3
Demand Forecasting	DemandForecasts	Sprint 4
Stockout Risk	DemandForecasts	Sprint 4
Intelligent Reordering	ReorderRecommendations	Sprint 4
Reporting	Multiple database tables	Sprint 5
Security	Users + Role-Based Access	Sprint 1 & 5

10. Sprint Prioritization

High Priority:

- Authentication and user roles
- Inventory management
- Supplier management
- Stock monitoring
- Inventory transactions
- Purchase orders

- Partial fulfillment

Medium Priority:

- Demand forecasting
- Stockout risk
- Intelligent reorder recommendations
- Dashboard and reporting

Finalization:

- System integration
- Testing
- Security validation
- Backup verification
- Documentation

11. Definition of Done

- Required functionality has been implemented.
- Required database entities and relationships are created.
- Primary and foreign-key relationships are working.
- Input validation has been implemented.
- Role-based access has been verified.
- The feature has been tested.
- No critical defects remain.
- The feature works with related modules.
- The feature meets its acceptance criteria.
- The code is integrated into the main project.

12. Sprint Review

Sprint	Demonstration
Sprint 1	Login, registration, user roles and database connection
Sprint 2	Add/update/view supplies and supplier management
Sprint 3	Low-stock detection, stock transactions, purchase orders and partial fulfillment
Sprint 4	Historical consumption, demand forecast, stockout risk and reorder recommendation
Sprint 5	Complete integrated system, dashboard, reports, security and testing

13. Sprint Retrospective

At the end of every sprint, the team will review:

- What went well?
- What did not go well?
- What technical difficulties occurred?
- Were any tasks underestimated?

- What should be improved in the next sprint?
- Are additional tests or changes required?

14. Team Collaboration

Responsibility Area	Main Work
Backend Development	ASP.NET Core, business logic and application services
Database Development	SQL Server, Entity Framework Core, relationships and constraints
Frontend Development	Razor views, forms and dashboards
Authentication	Identity, roles and authorization
Inventory Module	Supplies, stock and transactions
Procurement Module	Suppliers and purchase orders
AI/ML Module	Demand forecasting, stockout risk and reorder recommendations
Testing & Documentation	Test cases, validation and project documentation

15. Database and Agile Integration

The database design directly supports the Agile development plan. The major development sequence follows the dependencies between the database entities and functional requirements.

Development Phase	Database Component	Purpose
Phase 1	Users	Authentication and authorization
Phase 2	Suppliers + LabSupplies	Inventory management
Phase 3	InventoryTransactions	Historical stock movement
Phase 4	PurchaseOrders + PurchaseOrderItems	Procurement and partial fulfillment
Phase 5	DemandForecasts	Demand prediction
Phase 6	ReorderRecommendations	Intelligent replenishment

16. Data Integrity During Development

The database constraints will be tested during the relevant sprints.

- Primary Key constraints ensure unique records.
- Foreign Key constraints maintain valid relationships.
- NOT NULL constraints prevent missing required values.
- UNIQUE constraints prevent duplicate values where required.
- CHECK constraints prevent invalid quantity values.

Examples:

- QuantityOnHand >= 0
- QuantityOrdered > 0
- QuantityReceived >= 0

17. Expected Final Product

- User and role management.
- Laboratory supply management.

- Supplier management.
- Stock-level monitoring.
- Inventory transaction history.
- Purchase-order creation and tracking.
- Partial order fulfillment.
- Historical inventory data.
- Demand forecasting.
- Stockout risk identification.
- Intelligent reorder recommendations.
- Dashboard and reporting information.